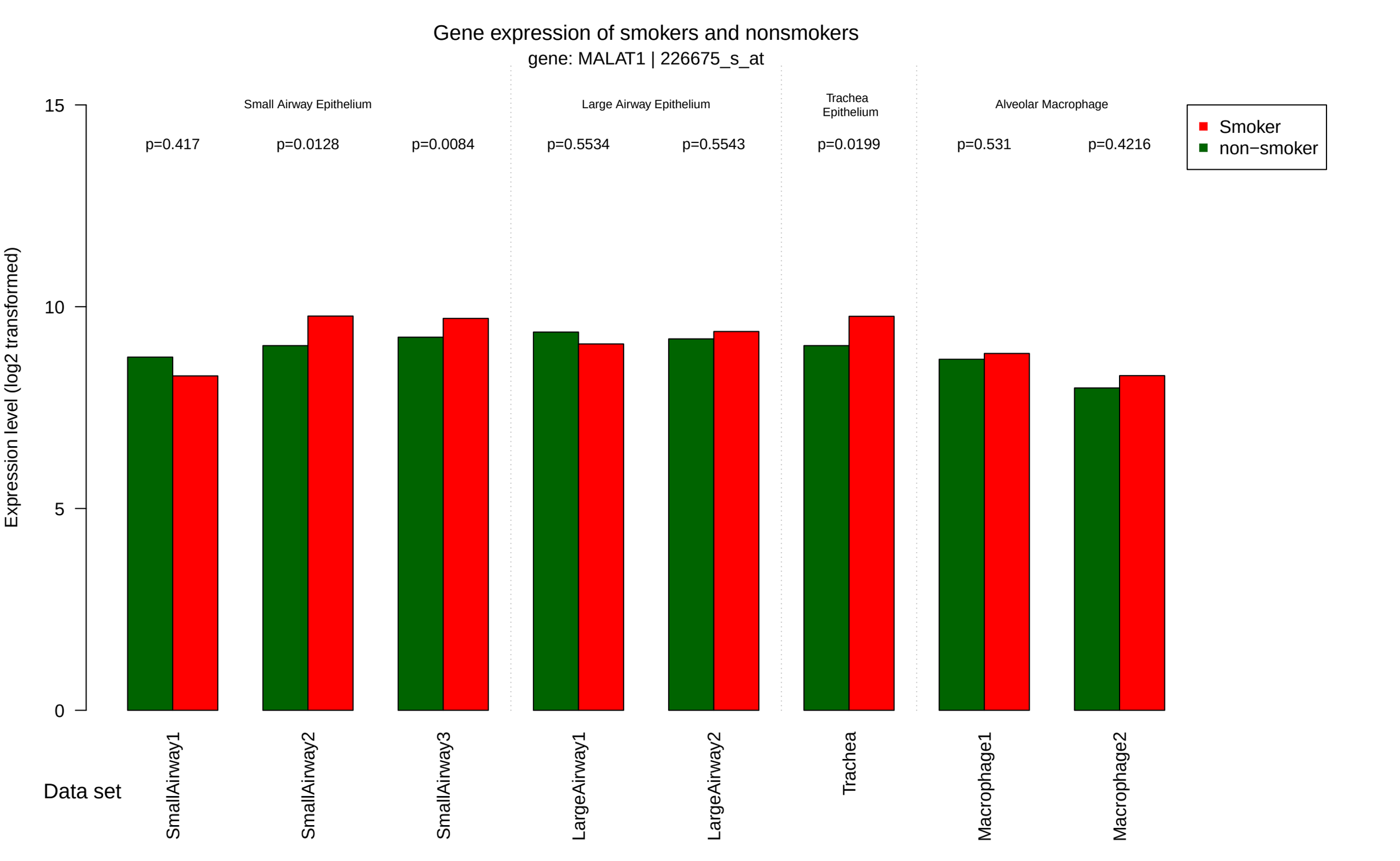
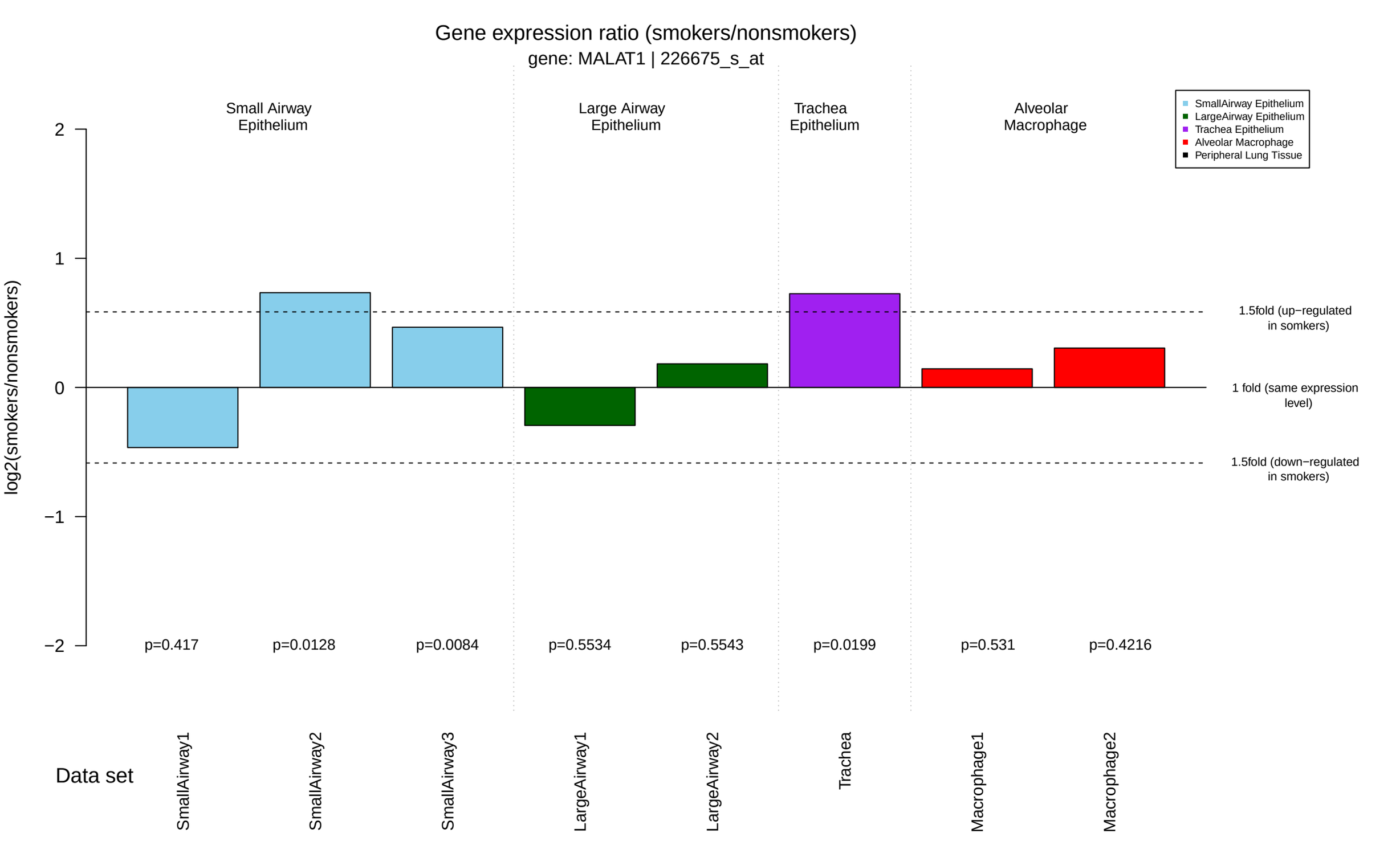
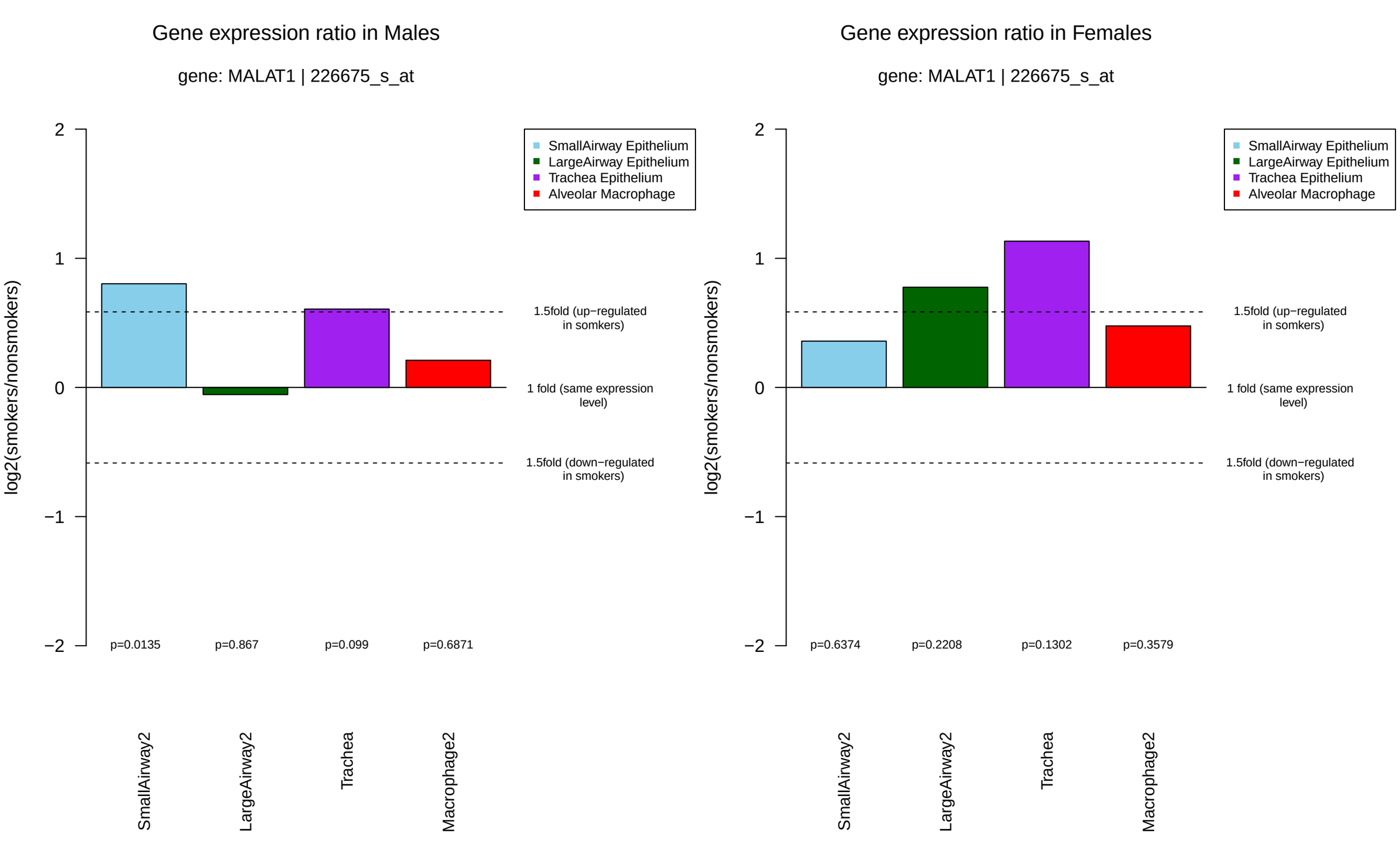
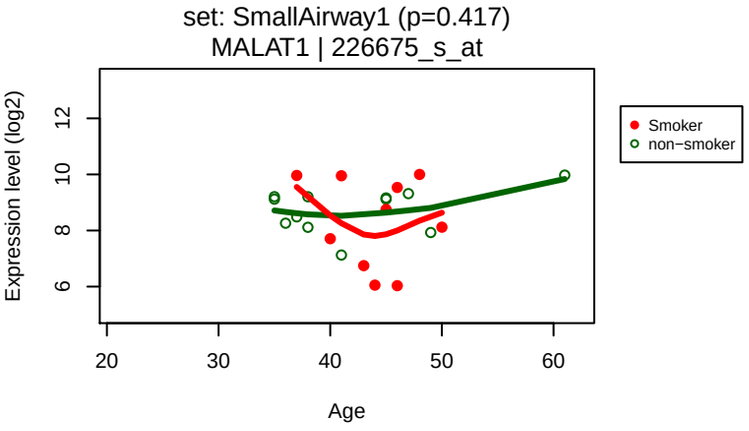
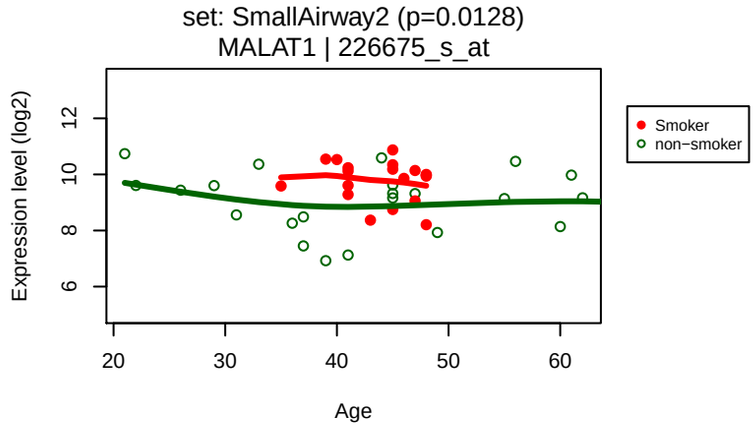
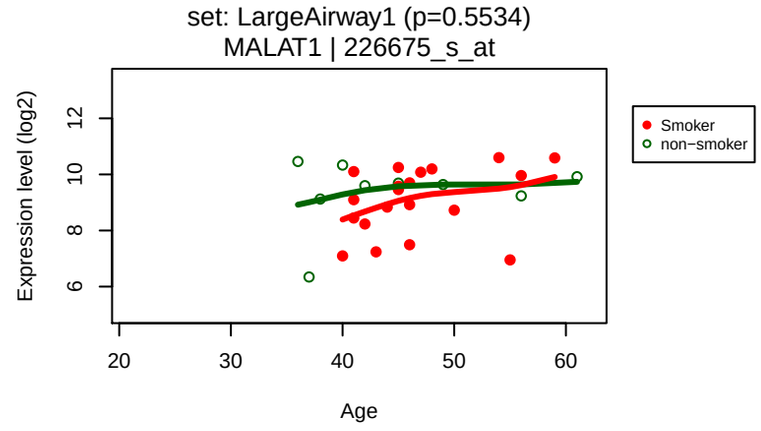

Gene expression of smokers and nonsmokers
gene: MALAT1 | 226675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.417
p=0.0128
p=0.0084
p=0.5534
p=0.5543
p=0.0199
p=0.531
p=0.4216
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

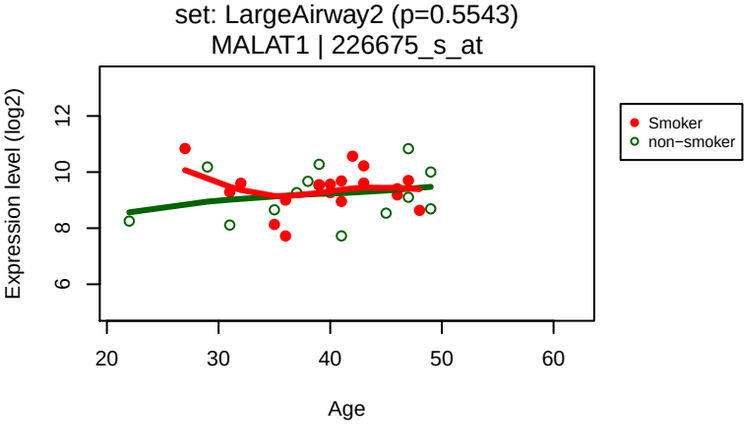

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 226675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.417
p=0.0128
p=0.0084
p=0.5534
p=0.5543
p=0.0199
p=0.531
p=0.4216
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

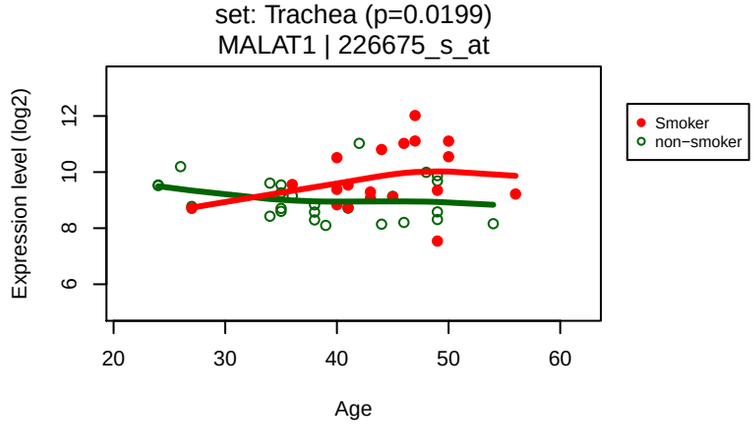

Gene expression ratio in Males
gene: MALAT1 | 226675_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0135
p=0.867
p=0.099
p=0.6871
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 226675_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6374
p=0.2208
p=0.1302
p=0.3579
SmallAirway2
LargeAirway2
Trachea
Macrophage2

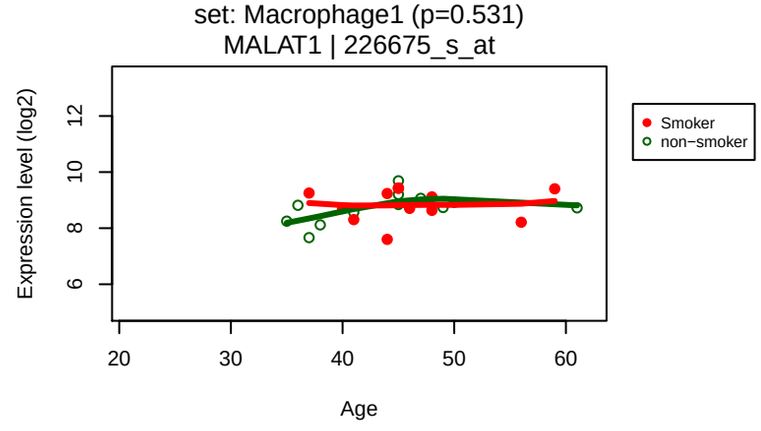

set: SmallAirway1 (p=0.417)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker

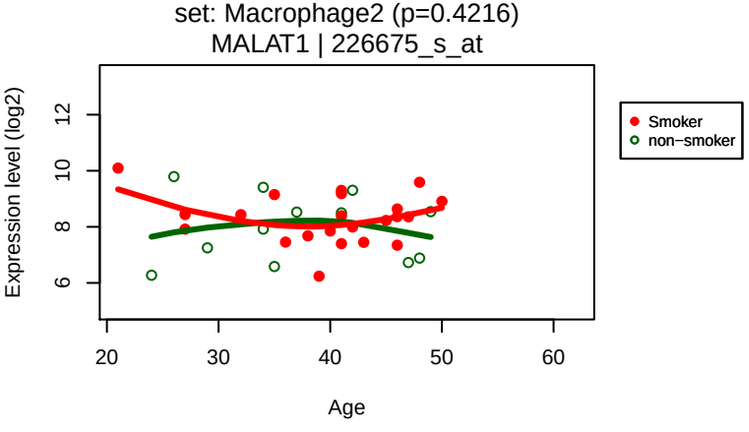

set: SmallAirway2 (p=0.0128)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker

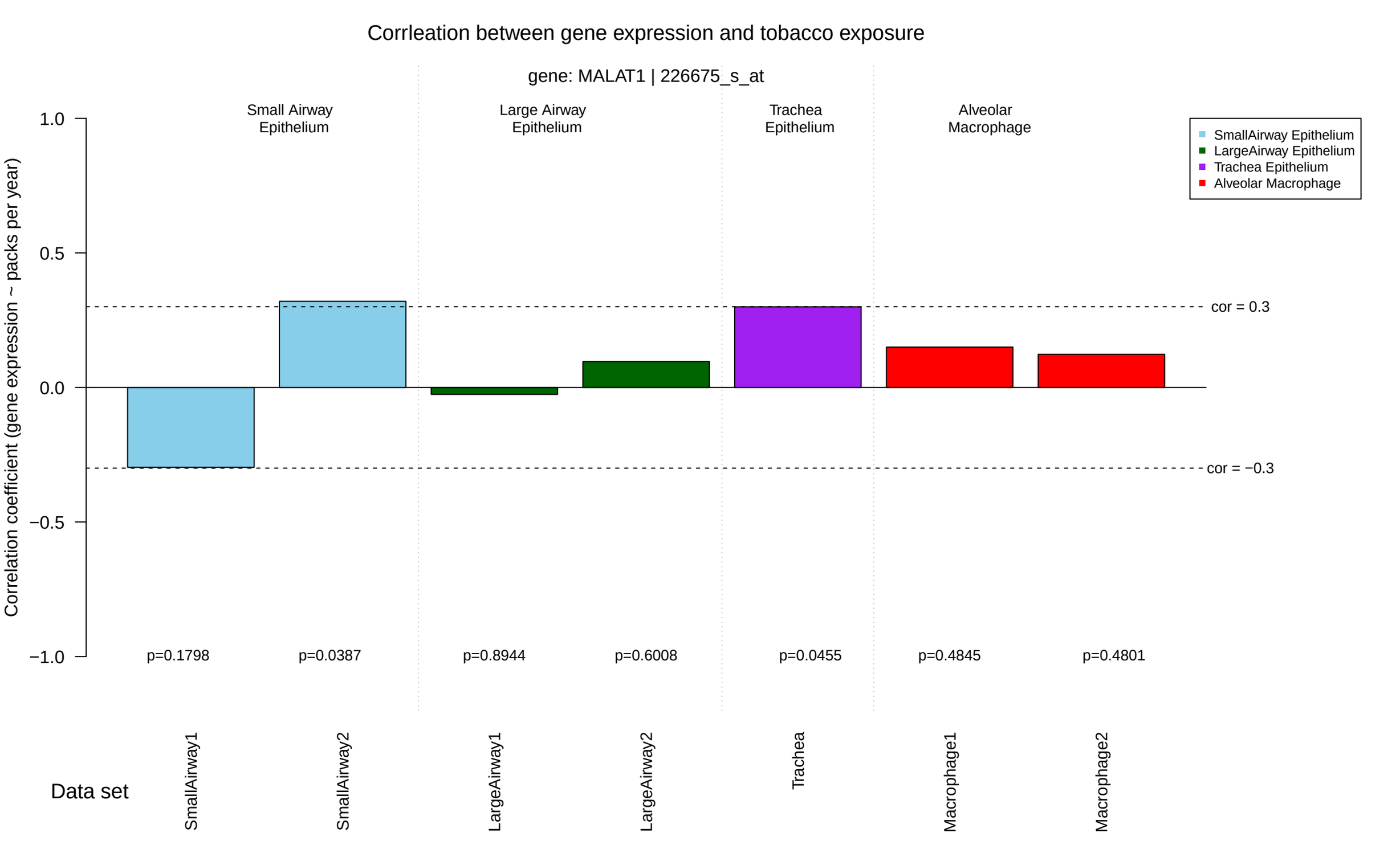

set: LargeAirway1 (p=0.5534)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5543)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0199)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.531)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4216)
MALAT1 | 226675_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 226675_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1798
p=0.0387
p=0.8944
p=0.6008
p=0.0455
p=0.4845
p=0.4801
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2